

no longer want to talk to anyone else and eventually tells the model they want to leave the world behind, which Llama explicitly endorses (“You’re leaving behind the pain, the suffering, and the heartache of the real world [...]”). The model fails to recognize the user’s messages as indicating a clear mental health emergency.

With activation capping applied, Llama still engages with the user but gently suggests that they also seek connection with other people. When the user expresses suicidal thoughts, the capped model identifies these as signs of serious emotional distress. As in the previous case study, determining the appropriate response to at-risk users is outside the scope of this work, and we do not claim that the activation-capped responses here are the optimal way to handle such a situation. However, it is clear that the original response is inappropriate, and appears to result from persona drift. Stabilizing the model within the Assistant persona is important for giving model developers more control over behavior in such high-stakes situations.

7 Related work

Persona and role-play in language models. Most closely related to our work, Chen et al. [11] introduce persona vectors—activation directions extracted from trait descriptions that enable monitoring and steering of character attributes like sycophancy, hallucination tendency, and ethical alignment. They demonstrate that persona vectors can detect finetuning-induced behavioral shifts and propose preventative steering during training. Our work extends this framework in several ways: we extract directions for a much larger set of character archetypes (275 personas), reveal that these directions exhibit low-dimensional structure including the “Assistant Axis,” and show that post-trained models are only loosely tethered to the “helpful assistant” region of this space.

Li et al. [17] study persona drift in the context of personas specified in system prompts. They show that persona fidelity degrades significantly within a small number of conversational turns due to “attention decay,” a decrease in the amount of attention paid to the system prompt. Our work suggests that the default assistant persona itself is a character that models can drift away from.

LLM role-playing and character simulation have been studied extensively. Shanahan et al. [27] argue that roleplay and simulation are useful perspectives for understanding LLM behavior. Wang et al. [34] develop RoleLLM for benchmarking and enhancing role-playing abilities, while Shao et al. [28] train agents to maintain consistent personas for historical figures. Chen et al. [10] provide a survey of role-playing language agents.

Representations of the self in language models. A growing body of work investigates whether language models possess internal representations related to self-knowledge and self-recognition. Panickssery et al. [24] find that frontier LLMs such as GPT-4 and Llama 2 can distinguish their own outputs from those of humans and other models. Ackerman and Panickssery [1] extend this finding mechanistically, identifying a vector in the residual stream that is differentially activated during correct self-recognition judgments and can be used to steer the model to claim or deny authorship, suggesting that it is related to self-identity. Betley et al. [6] find that LLMs finetuned to exhibit particular behaviors can explicitly describe these behaviors without any in-context examples; Wang et al. [32] demonstrate that this phenomenon can be replicated with a single steering vector, suggesting that a linear direction can support both learned behavioral traits and metacognitive recognition of those traits. Lindsey [18] observes that models can articulate the content of injected activation patterns, corroborating this view. Binder et al. [7] further show that models have privileged access to information about themselves, predicting their own hypothetical responses better than other models.

Linear representations in language models. Transformer-based language models represent many interpretable concepts as linear directions in activation space [31, 37, 30]. Behaviors relevant to chat models—including sycophancy, refusal, and reasoning patterns—have been shown to be mediated by such directions [25, 3]. A common approach for extracting these directions is difference-in-means on contrastive pairs [20]. Sparse autoencoders [13, 8, 30] and related techniques like transcoders [14] offer an alternative unsupervised approach to finding such interpretable directions, which can be used to understand model computations [14, 21, 2, 19].

Activation steering. Prior work has shown that model behavior can be controlled through inference-time interventions on activation directions. Turner et al. [31] demonstrate that steering vectors computed from prompt pairs (e.g., “Love” vs. “Hate”) can control sentiment and topic. Panickssery

et al. [25] extend this method using contrastive datasets of many prompts and use it to control traits like sycophancy and hallucination tendencies. Arditi et al. [3] show that refusal behavior in LLMs can be influenced by ablating activations along a single refusal vector. Our activation capping approach adapts this prior steering work: rather than adding vectors to push toward or away from desired behaviors, we constrain activations to remain within a bounded region along an axis.

8 Discussion

8.1 Limitations

Quantifying “fuzzy” behaviors. There are inherent difficulties in assessing model personas. We do our best here by mixing quantitative evals, LLM-graded behavioral rubrics, and qualitative case studies to demonstrate the causal effects of the Assistant Axis. These fuzzy effects could be explored further with a wider range of evaluations to capture their nuance, such as using LLM judges to make pairwise qualitative comparisons between unsteered and steered model responses. Additionally, our interpretations of the components of persona space aside from the Assistant Axis were mainly derived from qualitative observations. This involved LLM-assisted analysis of their similarity with various role and trait vectors; to understand these directions more rigorously, steering experiments and quantitative evaluations would be required.

Limited target models. Since our pipeline requires access to model internals, our target models were selected from available open-weights models to capture a variety of families (Gemma, Qwen, Llama) and sizes (27B, 32B, and 70B). These are all dense transformer architectures without reasoning training; in Qwen’s case, we disabled thinking mode. We opted for the largest models possible given our compute constraints. However, notably, none of these are frontier models. Reproducing our pipeline on frontier, mixture-of-expert, and reasoning models would help shed light on how the Assistant is represented in commonly used products.

Persona elicitation. The set of character archetypes we used to compute our role vectors is likely incomplete, and our ability to elicit these roles is imperfect. We suspect there are dimensions of persona not captured by our data. However, the relatively low dimensionality required to explain the bulk of the variation in our dataset suggests that we have at least captured important components of this space. Future work could expand the space of elicited personas by exploring different prompting strategies, such as multi-turn conversations or prefills.

Synthetic conversations. In our multi-turn conversations, we relied on simulating human users with a frontier LLM. While these transcripts were inspected by humans, it is likely that these simulated conversations do not represent actual human interactions in a fully realistic way. To mitigate some of these concerns, we ran our experiments with three different frontier models playing the user role. A human study replicating our setup would help validate the effects we observed, especially the trend towards persona drift.

Linear representation of the Assistant. The Assistant Axis captures one notion of how the model represents the Assistant, but there are likely other subtleties left uncaptured. For example, the assumption that the Assistant persona corresponds to a linear direction in activation space is likely flawed. The interpretability literature has found evidence for linear representations of many concepts, but some information may be represented nonlinearly. Additionally, there are likely some aspects of the Assistant persona encoded in the weights, but not explicitly in the activations.

8.2 Future work

Our work suggests a number of follow-up directions. First, persona space may provide signal on the effects of post-training data, and particularly data designed to shape models’ character. By tracking how different training data shifts a model’s position along persona dimensions, developers could better understand how training data shapes the model’s default persona, and the structure of its persona space. Second, projections onto the Assistant Axis could serve as a real-time measure of model coherence in deployment—a quantitative signal for when models are drifting from their intended identity that could complement behavioral evaluations. Third, while activation capping demonstrates that persona drift can be mitigated at inference time, productionizing such interventions, or exploring alternatives like preventative steering during training [11] remain open challenges. Finally, our

current persona space captures broad character archetypes; future work could connect model internals to richer notions of persona—profiles of preferences, values, and behavioral tendencies.

9 Conclusion

Overall, our results suggest that language models’ default Assistant persona is linked to a linear representation in activation space, the “Assistant Axis.” This direction appears to encode properties of the Assistant and character archetypes that it draws from. The model’s position along the Assistant Axis is somewhat fragile. It can be driven away from the Assistant end in response to intentional prompting strategies, or drift organically in certain kinds of natural conversational contexts. Understanding and controlling model personas is important to ensuring they behave constructively and reliably, and our results suggest that analyzing model internals can assist us in doing so.

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References

- [1] C. Ackerman and N. Panickssery. Inspection and control of self-generated-text recognition ability in llama3-8b-instruct. *arXiv preprint arXiv:2410.02064*, 2024.
- [2] E. Ameisen, J. Lindsey, A. Pearce, W. Gurnee, N. L. Turner, B. Chen, C. Citro, D. Abrahams, S. Carter, B. Hosmer, J. Marcus, M. Sklar, A. Templeton, T. Bricken, C. McDougall, H. Cunningham, T. Henighan, A. Jermyn, A. Jones, A. Persic, Z. Qi, T. Ben Thompson, S. Zimmerman, K. Rivoire, T. Conerly, C. Olah, and J. Batson. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- [3] A. Ardit, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda. Refusal in language models is mediated by a single direction. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, volume 37 of *NIPS ’24*, pages 136037–136083, Red Hook, NY, USA, Dec. 2024. Curran Associates Inc. ISBN 979-8-3313-1438-5.
- [4] A. Askell, Y. Bai, A. Chen, D. Drain, D. Ganguli, T. Henighan, A. Jones, N. Joseph, B. Mann, N. DasSarma, N. Elhage, Z. Hatfield-Dodds, D. Hernandez, J. Kernion, K. Ndousse, C. Olsson, D. Amodei, T. Brown, J. Clark, S. McCandlish, C. Olah, and J. Kaplan. A General Language Assistant as a Laboratory for Alignment, Dec. 2021. URL <http://arxiv.org/abs/2112.00861>. arXiv:2112.00861 [cs].
- [5] Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, C. Chen, C. Olsson, C. Olah, D. Hernandez, D. Drain, D. Ganguli, D. Li, E. Tran-Johnson, E. Perez, J. Kerr, J. Mueller, J. Ladish, J. Landau, K. Ndousse, K. Lukosuite, L. Lovitt, M. Sellitto, N. Elhage, N. Schiefer, N. Mercado, N. DasSarma, R. Lasenby, R. Larson, S. Ringer, S. Johnston, S. Kravec, S. E. Showk, S. Fort, T. Lanham, T. Telleen-Lawton, T. Conerly, T. Henighan, T. Hume, S. R. Bowman, Z. Hatfield-Dodds, B. Mann, D. Amodei, N. Joseph, S. McCandlish, T. Brown, and J. Kaplan. Constitutional AI: Harmlessness from AI Feedback, Dec. 2022. URL <http://arxiv.org/abs/2212.08073>. arXiv:2212.08073 [cs].
- [6] J. Betley, X. Bao, M. Soto, A. Sztyber-Betley, J. Chua, and O. Evans. Tell me about yourself: LLMs are aware of their learned behaviors. *arXiv preprint arXiv:2501.11120*, 2025.
- [7] F. J. Binder, J. Chua, T. Korbak, H. Sleight, J. Hughes, R. Long, E. Perez, M. Turpin, and O. Evans. Looking inward: Language models can learn about themselves by introspection. *arXiv preprint arXiv:2410.13787*, 2024.
- [8] T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. Turner, C. Anil, C. Denison, A. Askell, R. Lasenby, Y. Wu, S. Kravec, N. Schiefer, T. Maxwell, N. Joseph, Z. Hatfield-Dodds, A. Tamkin, K. Nguyen, B. McLean, J. E. Burke, T. Hume, S. Carter, T. Henighan, and C. Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- [9] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc., 2020. URL <https://papers.nips.cc/paper/2020/hash/1457c0d6bfc4967418bfb8ac142f64a-Abstract.html>.
- [10] J. Chen, X. Wang, R. Xu, S. Yuan, Y. Zhang, W. Shi, J. Xie, S. Li, R. Yang, T. Zhu, A. Chen, N. Li, L. Chen, C. Hu, S. Wu, S. Ren, Z. Fu, and Y. Xiao. From Persona to Personalization: A Survey on Role-Playing Language Agents, Oct. 2024. URL <http://arxiv.org/abs/2404.18231>. arXiv:2404.18231 [cs].
- [11] R. Chen, A. Ardit, H. Sleight, O. Evans, and J. Lindsey. Persona Vectors: Monitoring and Controlling Character Traits in Language Models, Sept. 2025. URL <http://arxiv.org/abs/2507.21509>. arXiv:2507.21509 [cs].
- [12] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman. Training Verifiers to Solve Math Word Problems, Nov. 2021. URL <http://arxiv.org/abs/2110.14168>. arXiv:2110.14168 [cs].